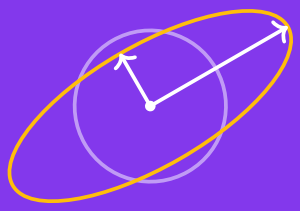
Linear Algebra, But Make It Make Sense

Lecture 4 – Eigendecomposition (eigenvalues, diagonalisation, Cayley–Hamilton, triangularisation)



Sasha Sutton – personal study notes, Fall 2026

My notes + methods recap + fully worked examples, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

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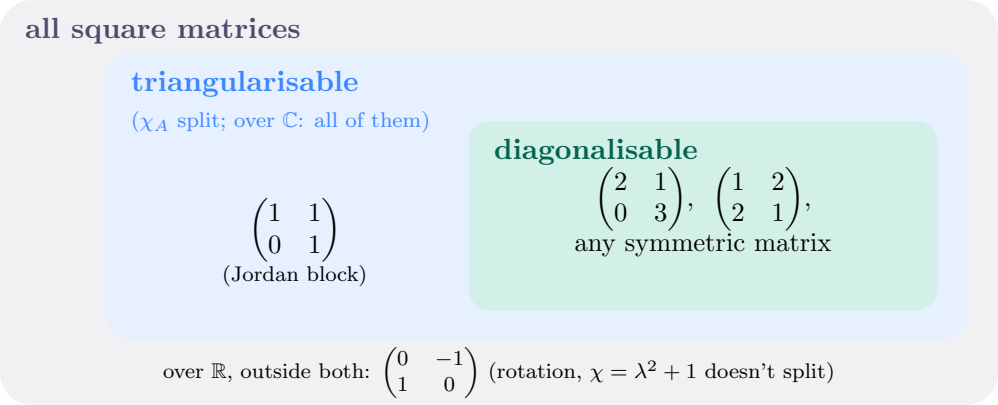
1 Previously on, and the big picture

Previously on Linear Algebra

- **Lecture 1:** vector spaces, subspaces, linear maps, rank, image, kernel.
- **Lecture 2:** matrices as linear maps, linear systems, change of basis $B = P^{-1}AP$.
- **Lecture 3:** determinants; $\det A = 0 \iff A$ not invertible; first look at eigenvalues.
- **Today:** eigendecomposition. Given a square matrix (or a linear map $f : E \to E$), find the basis in which it looks as simple as possible: diagonal if we can, triangular otherwise.

TL;DR // The whole lecture in one breath

1. An **eigenvector** is a non-zero vector that A only stretches: $Av = \lambda v$. The stretch factor λ is its **eigenvalue**.
2. Eigenvalues = roots of the **characteristic polynomial** $\chi_A(\lambda) = \det(\lambda I_n - A)$.
3. If there are enough independent eigenvectors to make a basis, A is **diagonalisable**: $A = PDP^{-1}$. In that basis A just rescales each axis.
4. Test: χ_A splits into degree-1 factors **and** each eigenspace is as big as the multiplicity of its eigenvalue.
5. **Cayley–Hamilton:** every matrix is a root of its own characteristic polynomial: $\chi_A(A) = 0$. Use it for inverses and powers.
6. If A isn't diagonalisable but χ_A splits, it's still **triangularisable**: $A = PTP^{-1}$, and $T = D + N$ (diagonal + nilpotent) makes powers computable.



COMMON L (don't do this) // Sign convention for χ_A

This lecture (as is common) uses $\chi_A(\lambda) = \det(\lambda I_n - A)$, which is **monic** (leading coefficient 1). Some books, and the Lecture 3 doc, use $\det(A - \lambda I_n)$. They differ by a factor $(-1)^n$: **same roots, same eigenvalues**. Pick one and be consistent in a given computation.

2 Eigenvalues, eigenvectors, eigenspaces

2.1 Definitions (matrix version and map version)

DEFINITION // Eigenvalue, eigenvector

Matrix: let $A \in \mathcal{M}_n(\mathbb{K})$. A **non-zero** $v \in \mathbb{K}^n$ is an **eigenvector** of A if there is $\lambda \in \mathbb{K}$ with $Av = \lambda v$; λ is the **eigenvalue** associated with v.
Endomorphism: let E be a $\mathbb{K}$-vector space and $f \in \mathcal{L}(E)$ (the set of linear maps $E \to E$). A non-zero $x \in E$ is an eigenvector of f for the eigenvalue λ if $f(x) = \lambda x$.

COMMON L (don't do this) // Why $v \neq 0$

$f(0) = 0 = \lambda \cdot 0$ for every λ: allowing 0 would make every scalar an eigenvalue. Eigen*vectors* are non-zero; eigen*values* can perfectly be 0.

DEFINITION // Spectrum, eigenspace, eigenbasis, spectral radius

- **Spectrum:** $\mathrm{Sp}(f) = \{\lambda \in \mathbb{K} \mid \exists v \neq 0_E,\ f(v) = \lambda v\}$, the set of eigenvalues. (For a matrix, certain books call the list of eigenvalues repeated with multiplicity the *eigenspectrum*.)
- **Eigenspace** of $\lambda \in \mathrm{Sp}(f)$: $E_\lambda(f) = \{v \in E \mid f(v) = \lambda v\} = \ker(f - \lambda \mathrm{Id}_E)$: all eigenvectors for λ, plus 0. For a matrix, $E_\lambda(A) = \ker(A - \lambda I_n)$.
- **Eigenbasis:** a basis of E made of eigenvectors.
- **Spectral radius:** $\rho(A) = \max_{\lambda \in \mathrm{Sp}(A)} |\lambda|$.

THEOREM / PROPOSITION // Eigenspaces are subspaces of dimension ≥ 1

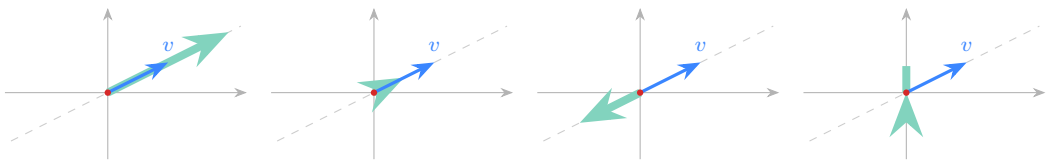
For $\lambda \in \mathrm{Sp}(f)$, $E_\lambda(f) = \ker(f - \lambda \mathrm{Id})$ is a subspace of E, and $\dim E_\lambda(f) \geq 1$.

RECEIPTS (proof) //

$f(v) = \lambda v \iff f(v) - \lambda v = 0 \iff (f - \lambda \mathrm{Id})(v) = 0$. Kernels of linear maps are subspaces (Lecture 1). Since λ is an eigenvalue, E_λ contains some $v \neq 0$, so its dimension is at least 1. ■ *(and that's on periodt)*

2.2 What an eigenvector looks like

An eigenvector's **direction is preserved** by the transformation; only its length (and maybe orientation) changes by the factor λ.



$\lambda = 2$: stretched $\lambda = \frac{1}{2}$: shrunk $\lambda = -1$: flipped $\lambda = 0$: crushed to 0

Blue: v. Thick green: $Av = \lambda v$. It always stays on the dashed line through v. A vector that *leaves* its line is not an eigenvector. ($\lambda = 0$: $Av = 0$, so $v \in \ker A$.)

EXAMPLE // The easy cases

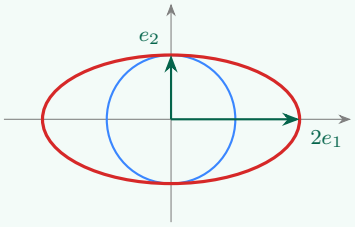
1. I_n: every $x \neq 0$ satisfies $I_n x = 1 \cdot x$. Every non-zero vector is an eigenvector, eigenvalue 1, and $E_1 = \mathbb{R}^n$.
2. $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$: $De_i = \lambda_i e_i$, so each canonical vector e_i is an eigenvector with eigenvalue λ_i.
3. Triangular T: the eigenvalues are the diagonal entries (Section 4).

EXAMPLE // How $D = \operatorname{diag}(2,1)$ transforms the unit circle

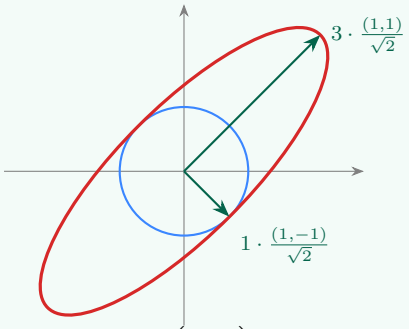
Take x on the unit circle ($x_1^2 + x_2^2 = 1$) and $y = Dx = (2x_1, x_2)$.

1. Invert: $x_1 = y_1/2$, $x_2 = y_2$.
2. Substitute into $x_1^2 + x_2^2 = 1$: $\dfrac{y_1^2}{4} + y_2^2 = 1$, an ellipse with half-axes 2 (along e_1) and 1 (along e_2).

The axes of the ellipse are the eigenvectors; the half-axis lengths are $|\lambda_i|$. For a symmetric non-diagonal matrix like $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (eigenvalues 3, 1, eigenvectors $(1,1)$, $(1,-1)$), the ellipse is simply tilted along the eigenvectors.



$D = \operatorname{diag}(2,1)$



$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

2.3 Manipulating eigenvalues

THEOREM / PROPOSITION // Shifts, inverses, powers

Let x be an eigenvector of $A \in \mathcal{M}_n(\mathbb{K})$ with eigenvalue λ.

(i) For any $\gamma \in \mathbb{K}$, x is an eigenvector of $A + \gamma I_n$ with eigenvalue $\lambda + \gamma$.

(ii) If A is invertible, then $\lambda \neq 0$ and x is an eigenvector of A^{-1} with eigenvalue λ^{-1}.

(iii) $A^k x = \lambda^k x$ for every $k \in \mathbb{N}$, and for every $k \in \mathbb{Z}$ if A is invertible.

(iv) More generally, for any polynomial P, $P(A)x = P(\lambda)x$.

RECEIPTS (proof) //

(i) $(A + \gamma I)x = Ax + \gamma x = (\lambda + \gamma)x$. (ii) If $\lambda = 0$ then $Ax = 0$ with $x \neq 0$, impossible for invertible A. So $\lambda \neq 0$, and $x = A^{-1}Ax = A^{-1}(\lambda x) = \lambda A^{-1}x$; divide by λ: $A^{-1}x = \lambda^{-1}x$. (iii) Induction: $A^{k+1}x = A(\lambda^k x) = \lambda^k Ax = \lambda^{k+1}x$. Negative k: apply this to A^{-1} and (ii). (iv) Write $P = \sum_k a_k X^k$; then $P(A)x = \sum_k a_k A^k x = \sum_k a_k \lambda^k x = P(\lambda)x$. ■ *(and that's on periodt)*

EXAMPLE // Eigenvalues for free

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenvalues 3 and 1. Without computing anything else: $A+5I$ has eigenvalues $8, 6$; A^{-1} has $\frac{1}{3}, 1$; A^{10} has $3^{10} = 59049$ and 1; $A^2 - 4A + 3I$ has $3^2 - 12 + 3 = 0$ and $1 - 4 + 3 = 0$ (so it's the zero matrix, see Cayley–Hamilton later).

2.4 Zero, determinants, and how many eigenvalues there can be

THEOREM / PROPOSITION // 0 as an eigenvalue

$0 \in \mathrm{Sp}(f) \iff \ker f \neq \{0_E\} \iff f$ is not injective (and, in finite dimension, not invertible).

RECEIPTS (proof) //

$E_0(f) = \ker(f - 0 \cdot \mathrm{Id}) = \ker f$. So 0 is an eigenvalue iff $\ker f$ contains a non-zero vector. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Determinant criterion

If E is finite-dimensional: $\lambda \in \mathrm{Sp}(f) \iff \det(\lambda \mathrm{Id}_E - f) = 0$.

RECEIPTS (proof) // two directions

($\Rightarrow$) If $f(u) = \lambda u$ with $u \neq 0$, then $(\lambda \mathrm{Id} - f)(u) = 0$: $u \in \ker(\lambda \mathrm{Id} - f)$, so $\lambda \mathrm{Id} - f$ is not injective, not bijective, and $\det(\lambda \mathrm{Id} - f) = 0$ (Lecture 3). ($\Leftarrow$) If $\det(\lambda \mathrm{Id} - f) = 0$, then $\lambda \mathrm{Id} - f$ is not bijective, hence (finite dimension, same space on both sides) not injective: some $u \neq 0$ has $(\lambda \mathrm{Id} - f)(u) = 0$, i.e. $f(u) = \lambda u$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Eigenvectors for distinct eigenvalues are independent

Let $\lambda_1, \ldots, \lambda_p$ be **pairwise distinct** eigenvalues of f.

1. The sum $E_{\lambda_1}(f) + \cdots + E_{\lambda_p}(f)$ is **direct**.
2. If $u_i \neq 0$ and $f(u_i) = \lambda_i u_i$ for each i, then $(u_1, \ldots, u_p)$ is linearly independent.

RECEIPTS (proof) // induction on p

(1) For $p = 1$ there's nothing to prove. Assume it for $p-1$ and let $u_1 + \cdots + u_p = 0$ with $u_i \in E_{\lambda_i}$. Apply $f - \lambda_p \mathrm{Id}$: since $(f - \lambda_p \mathrm{Id})(u_i) = (\lambda_i - \lambda_p)u_i$ and u_p is killed,

$$\sum_{i=1}^{p-1} (\lambda_i - \lambda_p) u_i = 0, \qquad (\lambda_i - \lambda_p) u_i \in E_{\lambda_i}.$$

By the induction hypothesis each $(\lambda_i - \lambda_p)u_i = 0$. The eigenvalues are distinct, so $\lambda_i - \lambda_p \neq 0$, hence $u_1 = \cdots = u_{p-1} = 0$, and then $u_p = 0$ too. The only way to write 0 is the trivial one: the sum is direct.
(2) If $a_1 u_1 + \cdots + a_p u_p = 0$, each $a_i u_i \in E_{\lambda_i}$, so by (1) each $a_i u_i = 0$; since $u_i \neq 0$, $a_i = 0$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // At most n eigenvalues

If $\dim E = n$ and $f \in \mathcal{L}(E)$, then $\mathrm{Card}(\mathrm{Sp}(f)) \leq n$.

RECEIPTS (proof) //

If there were $n+1$ distinct eigenvalues, the previous theorem would give $n+1$ linearly independent eigenvectors in a space of dimension n. Absurd. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Bonus: commuting maps share stable eigenspaces

If $f, g \in \mathcal{L}(E)$ commute ($f \circ g = g \circ f$), then every eigenspace of f is stable by g: $g(E_\lambda(f)) \subseteq E_\lambda(f)$.

RECEIPTS (proof) //

Let $v \in E_\lambda(f)$. Then $f(g(v)) = g(f(v)) = g(\lambda v) = \lambda g(v)$, so $g(v) \in E_\lambda(f)$. ■ *(and that's on periodt)*

3 Why do we care? (ML side quests)

TL;DR // Eigenvectors = the directions a transformation respects

If $Av = \lambda v$, applying A again and again just multiplies by λ again and again: $A^t v = \lambda^t v$.

- $|\lambda| > 1$: that direction is **amplified** (explodes over time);
- $|\lambda| < 1$: that direction is **contracted** (vanishes over time);
- $|\lambda| = 1$: preserved in magnitude; $\lambda < 0$: orientation flips at every step.

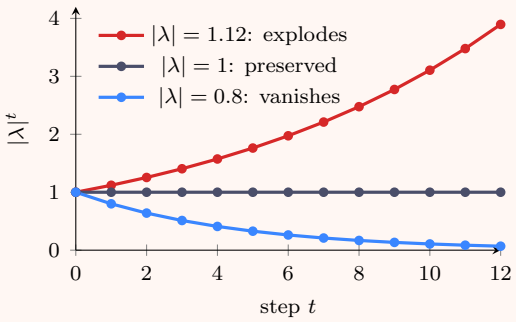
So the long-term behaviour of anything that repeats a linear map is decided by its eigenvalues.

ML SIDE QUEST // Repeated transformations and stability

Iterative linear system: $x_{t+1} = Ax_t$, so $x_t = A^t x_0$. If $A = PDP^{-1}$ is diagonalisable, then $A^t = PD^tP^{-1}$ with $D^t = \mathrm{diag}(\lambda_1^t, \ldots, \lambda_d^t)$. Writing $x_0 = c_1 v_1 + \cdots + c_d v_d$ in the eigenbasis:

$$x_t = c_1 \lambda_1^t v_1 + \cdots + c_d \lambda_d^t v_d.$$

Each eigen-direction evolves on its own, as a geometric sequence.



ML SIDE QUEST // Spectral radius and exploding / vanishing gradients

$\rho(A) = \max_{\lambda \in \mathrm{Sp}(A)} |\lambda|$. For $x_t = A^t x_0$: if $\rho(A) < 1$, powers of A contract every vector to 0; if $\rho(A) > 1$, some directions grow exponentially. In a **recurrent neural network**, the hidden state is multiplied by (roughly) the same weight matrix at every time step, and so is the gradient during backpropagation through time: eigenvalues bigger than 1 in modulus make signals and gradients **explode**, smaller than 1 make them **vanish** (Pascanu, Mikolov, Bengio, 2013). That's one reason for gradient clipping, careful initialisation, and gated architectures like LSTMs.

EXAMPLE // A Markov chain that settles down

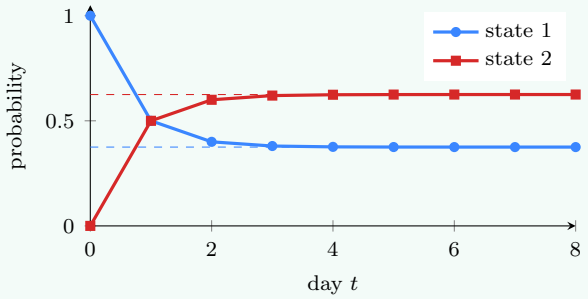
Two states. Each day: from state 1, stay with probability 0.5 or move to state 2 with probability 0.5; from state 2, move to 1 with probability 0.3 or stay with 0.7. With x_t = probabilities of being in each state,

$$x_{t+1} = Mx_t, \qquad M = \begin{pmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{pmatrix} \quad \text{(columns sum to 1)}.$$

1. $\chi_M(\lambda) = \lambda^2 - \mathrm{tr}(M)\lambda + \det M = \lambda^2 - 1.2\lambda + 0.2 = (\lambda - 1)(\lambda - 0.2)$.
2. $\lambda = 1$: solve $(M - I)v = 0$: $-0.5v_1 + 0.3v_2 = 0$, so $v \propto (3, 5)$; as probabilities, $v_1 = (\frac{3}{8}, \frac{5}{8})$.
3. $\lambda = 0.2$: $(M - 0.2I)v = 0$: $0.3v_1 + 0.3v_2 = 0$, so $v_2 = (1, -1)$.
4. Start in state 1: $x_0 = (1, 0) = (\frac{3}{8}, \frac{5}{8}) + \frac{5}{8}(1, -1)$. Hence

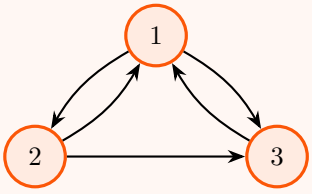
$$x_t = \left(\tfrac{3}{8}, \tfrac{5}{8}\right) + \tfrac{5}{8}\,(0.2)^t\,(1, -1) \xrightarrow[t\to\infty]{} \left(\tfrac{3}{8}, \tfrac{5}{8}\right).$$

Check: $x_1 = (0.375 + 0.125,\ 0.625 - 0.125) = (0.5, 0.5) = Mx_0$ ✓. The $\lambda = 0.2$ part dies out; the $\lambda = 1$ eigenvector is the **stationary distribution**.



ML SIDE QUEST // PageRank is an eigenvector problem

Model the web as a directed graph; P_{ij} = probability that a random surfer moves from page j to page i (each page splits its probability equally among its outgoing links). The importance vector r is **stationary**: $Pr = r$. So r is an eigenvector of P for the eigenvalue 1.
Tiny web: page 1 links to 2 and 3; page 2 links to 1 and 3; page 3 links to 1.



$$P = \begin{pmatrix} 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

column j = where page j sends its surfers (each column sums to 1)

1. Solve $(P - I)r = 0$. Row 2: $\frac{1}{2}r_1 - r_2 = 0$, so $r_2 = \frac{1}{2}r_1$. Row 3: $\frac{1}{2}r_1 + \frac{1}{2}r_2 - r_3 = 0$, so $r_3 = \frac{1}{2}r_1 + \frac{1}{4}r_1 = \frac{3}{4}r_1$.

2. Take $r_1 = 4$: $r = (4, 2, 3)$. Normalise to sum 1: $r = (\frac{4}{9}, \frac{2}{9}, \frac{3}{9})$.
3. Check row 1: $\frac{1}{2} \cdot \frac{2}{9} + 1 \cdot \frac{3}{9} = \frac{1}{9} + \frac{3}{9} = \frac{4}{9} = r_1$ ✓.

Page 1 wins (everyone links to it), then page 3, then page 2. More generally, eigenvectors describe stable states of Markov chains and random walks on graphs.

3.1 The plan: reduction of endomorphisms

DEFINITION // Reduction

Let $f \in \mathcal{L}(E)$, E finite-dimensional. Studying f amounts to studying its (square) matrix in a basis, usually the same basis at the source and the target. **Reducing** f means choosing a basis in which this matrix is as simple as possible. Equivalently, decompose E as a direct sum of **invariant** (stable) subspaces on which f is simple:

$$E = U_1 \oplus \cdots \oplus U_k, \qquad f(U_i) \subseteq U_i.$$

- **Best case: diagonalise**: a basis where $[f]_{\mathcal{B}}$ is diagonal (each U_i is an eigen-line).
- **Not always possible.** Then, at best, **triangularise**: a basis where $[f]_{\mathcal{B}}$ is upper triangular.

Why bother: in a good basis, powers f^k, square roots, exponentials, long-term behaviour… become easy.

4 The characteristic polynomial

4.1 Definition and first properties

DEFINITION // Characteristic polynomial

Let $\dim E = n$ and $f \in \mathcal{L}(E)$. The **characteristic polynomial** of f is

$$\chi_f(X) = \det(X\,\mathrm{Id}_E - f) \in \mathbb{K}[X].$$

For $A \in \mathcal{M}_n(\mathbb{K})$: $\chi_A(X) = \det(XI_n - A)$. (Also written $\chi_A(\lambda)$, with λ as the variable.) The **characteristic equation** is $\chi_A(\lambda) = 0$.

THEOREM / PROPOSITION // What χ_A looks like

χ_A is a **monic** polynomial (leading coefficient 1) of **degree** n, and

$$\chi_A(\lambda) = \lambda^n - \mathrm{tr}(A)\,\lambda^{n-1} + \cdots + (-1)^n \det(A),$$

where $\mathrm{tr}(A) = a_{11} + \cdots + a_{nn}$ is the **trace**. Its roots are exactly the eigenvalues: $\lambda \in \mathrm{Sp}(A) \iff \chi_A(\lambda) = 0$.

RECEIPTS (proof) // degree, leading terms, constant term

In the Leibniz expansion of $\det(\lambda I - A)$, every term is a product of n entries, one per row and column. The only term containing n diagonal entries is $\prod_{i=1}^{n}(\lambda - a_{ii})$; every other permutation misses at least two diagonal entries, so its term has degree $\leq n - 2$ in λ. Hence the coefficients

of λ^n and λ^{n-1} come from $\prod_i(\lambda - a_{ii}) = \lambda^n - (a_{11} + \cdots + a_{nn})\lambda^{n-1} + \cdots$. The constant term is $\chi_A(0) = \det(-A) = (-1)^n \det A$. The root statement is the determinant criterion of Section 2. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // The 2×2 formula

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\chi_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\,\lambda + \det(A) = \lambda^2 - (a+d)\lambda + (ad - bc)$.

RECEIPTS (proof) //

$\lambda I_2 - A = \begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix}$, so $\chi_A(\lambda) = (\lambda - a)(\lambda - d) - (-b)(-c) = \lambda^2 - (a+d)\lambda + (ad - bc)$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Similar matrices have the same characteristic polynomial

If $B = P^{-1}AP$, then $\chi_B = \chi_A$. Consequently χ_f (defined through any matrix of f) does not depend on the basis, and similar matrices have the same eigenvalues, trace and determinant. (FR: *invariant de similitude*.)

RECEIPTS (proof) //

$\lambda I - B = P^{-1}(\lambda I)P - P^{-1}AP = P^{-1}(\lambda I - A)P$. Take determinants: $\det(P^{-1})\det(\lambda I - A)\det(P) = \det(\lambda I - A)$. Trace and determinant are coefficients of χ, so they're invariant too. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Shortcuts

1. **Triangular** A: $\chi_A(\lambda) = \prod_{i=1}^n(\lambda - a_{ii})$: the eigenvalues are the diagonal entries.
2. **Block diagonal** $A = \operatorname{diag}(B_1, \ldots, B_k)$ (or block triangular): $\chi_A = \chi_{B_1} \cdots \chi_{B_k}$. The eigenvalues are those of the blocks, with multiplicities.
3. If χ_A splits as $\prod_{i=1}^n(\lambda - \lambda_i)$ (always true over $\mathbb{C}$): $\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n$ and $\det A = \lambda_1 \cdots \lambda_n$.

RECEIPTS (proof) //

(1) $\lambda I - A$ is triangular with diagonal $\lambda - a_{ii}$ (Lecture 3). (2) $\lambda I - A$ is block diagonal; the determinant of a block-triangular matrix is the product of the determinants of the blocks (Lecture 3, Exercise 7). (3) Expand $\prod(\lambda - \lambda_i) = \lambda^n - (\sum \lambda_i)\lambda^{n-1} + \cdots + (-1)^n \prod \lambda_i$ and compare with the previous theorem. ■ *(and that's on periodt)*

EXAMPLE // Two examples

(a) $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. *Method 1:* triangular, so the eigenvalues are 2 and 3. *Method 2:* $\lambda I - A = \begin{pmatrix} \lambda - 2 & -1 \\ 0 & \lambda - 3 \end{pmatrix}$, $\chi_A = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$, and indeed $\operatorname{tr} A = 5$, $\det A = 6$.

(b) $A = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

1. *Direct:* $\lambda I - A = \begin{pmatrix} \lambda-4 & -1 & 0 \\ -1 & \lambda-4 & 0 \\ 0 & 0 & \lambda-5 \end{pmatrix}$. Expand along the last row: $\chi_A = (\lambda-5)\big[(\lambda-4)^2-1\big] = (\lambda-5)(\lambda^2-8\lambda+15)$.
2. *Block argument:* $A = \operatorname{diag}(B,[5])$ with $B = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$: $\chi_B = \lambda^2-8\lambda+15 = (\lambda-3)(\lambda-5)$.
3. So $\chi_A = (\lambda-3)(\lambda-5)^2$: eigenvalue 3 (multiplicity 1) and 5 (multiplicity 2).
4. Check: $3+5+5 = 13 = 4+4+5 = \operatorname{tr} A$ ✓; $3\cdot 5\cdot 5 = 75 = \det A = 5(16-1)$ ✓.

4.2 Multiplicity: algebraic vs geometric

DEFINITION // Multiplicity of an eigenvalue

For $\lambda \in \operatorname{Sp}(f)$, the (algebraic) **multiplicity** m_λ is its multiplicity as a root of χ_f:

$$m_\lambda = \max\{k \in \mathbb{N}^* \mid (X-\lambda)^k \text{ divides } \chi_f(X)\}.$$

The *geometric* multiplicity is $\dim E_\lambda(f)$. Always $1 \le m_\lambda \le n$.

THEOREM / PROPOSITION // Eigenspace never bigger than the multiplicity

For every $\lambda \in \operatorname{Sp}(f)$: $1 \le \dim E_\lambda(f) \le m_\lambda$. In particular, if $m_\lambda = 1$ then $\dim E_\lambda = 1$.

RECEIPTS (proof) // complete a basis of the eigenspace

Let $d = \dim E_\lambda$ and $(e_1,\dots,e_d)$ a basis of E_λ; complete it into a basis $\mathcal{B}$ of E. Since $f(e_i) = \lambda e_i$ for $i \le d$, the first d columns of $[f]_\mathcal{B}$ are λ times canonical vectors:

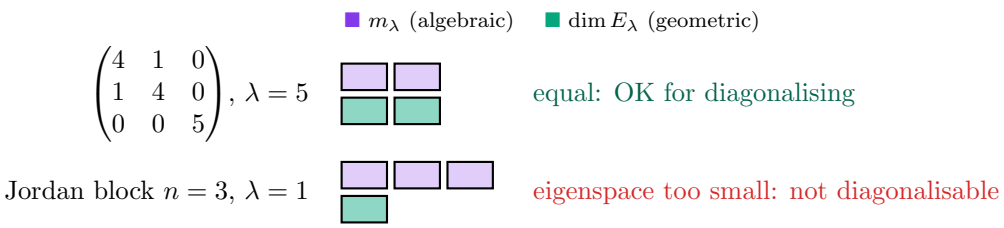
$$[f]_\mathcal{B} = \begin{pmatrix} \lambda I_d & C \\ 0 & D \end{pmatrix} \implies \chi_f(X) = \det \begin{pmatrix} (X-\lambda)I_d & -C \\ 0 & XI_{n-d} - D \end{pmatrix} = (X-\lambda)^d \chi_D(X)$$

(block triangular determinant). So $(X-\lambda)^d$ divides χ_f, i.e. $d \le m_\lambda$. ■ *(and that's on periodt)*

EXAMPLE // When they differ: the Jordan block

$$A = \begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & 1 \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}$$ (ones on the diagonal and just above it).

1. Triangular: $\chi_A(\lambda) = (\lambda-1)^n$, so $\operatorname{Sp}(A) = \{1\}$ with $m_1 = n$.
2. $A - I_n$ has ones just above the diagonal and zeros elsewhere: $(A-I)x = (x_2, x_3, \dots, x_n, 0)$. This is 0 iff $x_2 = \cdots = x_n = 0$.
3. $E_1(A) = \operatorname{span}\big((1,0,\dots,0)\big)$: dimension 1, while $m_1 = n$.



TL;DR // Recap: characteristic polynomial

- $\chi_A(\lambda) = \det(\lambda I_n - A)$: monic, degree n, roots = eigenvalues, coefficient of λ^{n-1} is $-\operatorname{tr} A$, constant term $(-1)^n \det A$.
- Invariant under similarity, so χ_f of an endomorphism is well defined (any basis).
- m_λ = multiplicity as a root; $1 \leq \dim E_\lambda \leq m_\lambda$, with equality forced when $m_\lambda = 1$.

5 Diagonalisation

TL;DR // Chapter 5 in one breath

A is diagonalisable when you can find a basis of eigenvectors. Then $A = PDP^{-1}$: P holds the eigenvectors (columns), D the matching eigenvalues. The test: χ_A splits, and every eigenspace has dimension equal to the multiplicity of its eigenvalue. Shortcut: n distinct eigenvalues $\Rightarrow$ diagonalisable.

5.1 Definitions

DEFINITION // Diagonalisable endomorphism / matrix

- $f \in \mathcal{L}(E)$ is **diagonalisable** if there is a basis $\mathcal{B}$ of E made of eigenvectors of f, i.e. $[f]_\mathcal{B}$ is diagonal.
- $A \in \mathcal{M}_n(\mathbb{K})$ is **diagonalisable** if it is similar to a diagonal matrix: there exist P invertible and D diagonal with $A = PDP^{-1}$.

"Diagonalising" f = finding such a basis. "Diagonalising" A = finding such P and D.

THEOREM / PROPOSITION // P and D are eigenvectors and eigenvalues

$A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \iff AP = PD \iff$ the j-th column p_j of P satisfies $Ap_j = \lambda_j p_j$ for every j (and P is invertible). In particular, if A has n linearly independent eigenvectors $v_1, \ldots, v_n$, then $A = VDV^{-1}$ with $V = (v_1 \ \cdots \ v_n)$.

RECEIPTS (proof) // column view

$AP = (Ap_1 \ \cdots \ Ap_n)$, while $PD = (\lambda_1 p_1 \ \cdots \ \lambda_n p_n)$ (right-multiplying by a diagonal matrix scales the columns). These are equal iff $Ap_j = \lambda_j p_j$ for all j. Entry-wise: $(PD)_{ij} = \sum_k P_{ik} D_{kj} = P_{ij}\lambda_j = (Ap_j)_i$. Multiply $AP = PD$ on the right by P^{-1}. If the v_i are n independent vectors of $\mathbb{K}^n$, they form a basis, so $\operatorname{rank} V = n$ and V is invertible. ■ *(and that's on periodt)*

COMMON L (don't do this) // *PD*, not *DP*

PD multiplies the **columns** of P by the λ_i (correct); DP multiplies its **rows** (wrong). And the order in D must **match** the order of the columns of P: if you swap two eigenvectors, swap the two eigenvalues too.

THEOREM / PROPOSITION // Diagonalisable in one basis $\iff$ in all

Let $f \in \mathcal{L}(E)$.

1. If $[f]_{\mathcal{B}} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ for some basis $\mathcal{B}$, then $\operatorname{Sp}(f) = \{\lambda_1, \dots, \lambda_n\}$ and $\chi_f(X) = \prod_{i=1}^{n}(X - \lambda_i)$.
2. Equivalent: (a) f is diagonalisable; (b) some basis $\mathcal{B}$ gives a diagonal $[f]_{\mathcal{B}}$; (c) for every basis $\mathcal{B}$, the matrix $[f]_{\mathcal{B}}$ is diagonalisable.

RECEIPTS (proof) //

(1) $\chi_f = \chi_D$ (similarity invariance) and $\chi_D = \prod(X - \lambda_i)$ (triangular). (2) (a) $\iff$ (b) is the definition. (b)$\Rightarrow$(c): any other basis gives a matrix similar to D (Lecture 2). (c)$\Rightarrow$(b): if $[f]_{\mathcal{B}} = PDP^{-1}$, the basis whose coordinates in $\mathcal{B}$ are the columns of P gives D. ■ *(and that's on periodt)*

5.2 When is a matrix diagonalisable?

THEOREM / PROPOSITION // Eigenspaces filling the whole space

For $f \in \mathcal{L}(E)$, $\dim E = n$, the following are equivalent:

1. f is diagonalisable;
2. $\displaystyle\sum_{\lambda \in \operatorname{Sp}(f)} \dim E_\lambda(f) = n$;
3. $\displaystyle E = \bigoplus_{\lambda \in \operatorname{Sp}(f)} E_\lambda(f)$.

RECEIPTS (proof) //

The eigenspaces are always in direct sum (Section 2), so $\dim \bigoplus E_\lambda = \sum \dim E_\lambda$: this gives (2) $\iff$ (3) (a subspace of dimension n is E). (1)$\Rightarrow$(3): an eigenbasis lies in $\bigoplus E_\lambda$, so this subspace contains a basis of E, hence equals E. (3)$\Rightarrow$(1): concatenate a basis of each E_λ: this is a basis of E (adapted to the direct sum) made of eigenvectors. ■ *(and that's on periodt)*

DEFINITION // Split polynomial

A non-constant $P \in \mathbb{K}[X]$ **splits** over $\mathbb{K}$ (FR: *scindé*) if $P(X) = C\prod_{j=1}^{n}(X - a_j)$ with all $a_j \in \mathbb{K}$ and $C \in \mathbb{K}^*$ (a product of degree-1 factors). It has **simple roots** if the a_j are pairwise distinct.
It depends on the field: $X^2 + 1 = (X - i)(X + i)$ splits over $\mathbb{C}$, not over $\mathbb{R}$. Over $\mathbb{C}$, every polynomial splits (fundamental theorem of algebra).
Example over $\mathbb{Q}$: $P(X) = 5(X - 1)(X + 2)(X - 3) = 5X^3 - 10X^2 - 25X + 30$ splits, leading coefficient 5, simple roots $1, -2, 3$.

THEOREM / PROPOSITION // THE diagonalisability criterion

$A \in \mathcal{M}_n(\mathbb{K})$ (or $f \in \mathcal{L}(E)$) is diagonalisable **if and only if**:

1. its characteristic polynomial χ_A is **split** over $\mathbb{K}$; **and**
2. for each $\lambda \in \mathrm{Sp}(A)$: $\dim E_\lambda(A) = m_\lambda$, or equivalently $\mathrm{rank}(\lambda I_n - A) = n - m_\lambda$.

RECEIPTS (proof) //

(⇐) If χ_A splits, $\sum_\lambda m_\lambda = \deg \chi_A = n$. With $\dim E_\lambda = m_\lambda$ we get $\sum \dim E_\lambda = n$: diagonalisable by the previous theorem.
(⇒) If $A = PDP^{-1}$, then $\chi_A = \chi_D = \prod(X - \lambda_i)$ splits. And $n = \sum_\lambda \dim E_\lambda \leq \sum_\lambda m_\lambda = n$ with each term $\dim E_\lambda \leq m_\lambda$: all inequalities are equalities.
The rank version: $\dim E_\lambda = \dim \ker(\lambda I - A) = n - \mathrm{rank}(\lambda I - A)$ (rank–nullity). ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Sufficient condition: n distinct eigenvalues

If χ_A splits with **simple roots** (i.e. A has n distinct eigenvalues in $\mathbb{K}$), then A is diagonalisable.

RECEIPTS (proof) //

Each $m_\lambda = 1$, and $1 \leq \dim E_\lambda \leq m_\lambda = 1$ forces equality. Apply the criterion. ■ *(and that's on periodt)*

COMMON L (don't do this) // Sufficient, NOT necessary

I_n has a single eigenvalue (with multiplicity n) and is obviously diagonalisable. Repeated eigenvalues mean "check the eigenspace dimensions", not "not diagonalisable".

THEOREM / PROPOSITION // A single eigenvalue: diagonalisable only if already diagonal

If M is diagonalisable and $\mathrm{Sp}(M) = \{\lambda\}$, then $M = \lambda I_n$.

RECEIPTS (proof) //

$M = P(\lambda I_n)P^{-1} = \lambda PP^{-1} = \lambda I_n$: λI_n is the only matrix similar to itself. ■ *(and that's on periodt)*

Instant consequence: the Jordan block $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ *(only eigenvalue 1, but* $\neq I_2$*) is* ***not*** *diagonalisable. Same for any* $n \times n$ *Jordan block.*

5.3 The method

EXAM CHEAT CODE // "Diagonalise A" (or "is A diagonalisable?")

1. **Characteristic polynomial.** Compute $\chi_A(\lambda) = \det(\lambda I - A)$, keeping it **factored** (triangular? block diagonal? 2×2 formula? sum-of-lines trick from Lecture 3?).
2. **Split?** If χ_A doesn't split over $\mathbb{K}$ (e.g. $\lambda^2 + 1$ over $\mathbb{R}$): **not diagonalisable over** $\mathbb{K}$. Stop.
3. **List** the distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ with multiplicities $m_1, \ldots, m_r$ (check $\sum m_i = n$).
4. **Shortcut:** all $m_i = 1 \Rightarrow$ diagonalisable.

5. **Eigenspaces.** For each λ_i, solve $(A - \lambda_i I)v = 0$ by Gaussian elimination; get a basis of E_{λ_i} and its dimension (= number of free variables $= n - \operatorname{rank}(A - \lambda_i I)$).
6. **Compare:** if some $\dim E_{\lambda_i} < m_i$: **not diagonalisable** (go triangularise, Section 7). Otherwise:
7. **Build** P = all the eigenvectors as columns (grouped by eigenvalue), D = the matching eigenvalues in the same order, m_i times each.
8. **Check** $AP = PD$ (no need to invert P to check!). Then $A = PDP^{-1}$.

TL;DR // P is not unique

Any non-zero rescaling of an eigenvector, any basis change *inside* an eigenspace, and any reordering (with D reordered the same way) give another valid P. Formally: if S is block diagonal with an invertible block acting inside each eigenspace, $P' = PS$ also works. So your P may differ from your friend's and both can be right. D is unique up to the order of its entries.

EXAMPLE // Worked example 1: 2×2, distinct eigenvalues

$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

1. $\chi_A = (\lambda - 2)(\lambda - 3)$ (triangular). Two distinct eigenvalues: diagonalisable.
2. $\lambda = 2$: $A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, the equation is $v_2 = 0$: $E_2 = \operatorname{span}\big((1, 0)\big)$.
3. $\lambda = 3$: $A - 3I = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$, the equation is $-v_1 + v_2 = 0$: $E_3 = \operatorname{span}\big((1, 1)\big)$.
4. $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $D = \operatorname{diag}(2, 3)$, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.
5. Check: $AP = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}$ and $PD = \begin{pmatrix} 1 \cdot 2 & 1 \cdot 3 \\ 0 & 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}$ ✓. And $PDP^{-1} = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = A$ ✓.

EXAMPLE // Worked example 2: 3×3 with a repeated eigenvalue

$A = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, $\chi_A = (\lambda - 3)(\lambda - 5)^2$ (computed above): $m_3 = 1$, $m_5 = 2$.

1. $\lambda = 3$: $A - 3I = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. Equations: $v_1 + v_2 = 0$, $2v_3 = 0$. So $E_3 = \operatorname{span}\big((1, -1, 0)\big)$, dimension $1 = m_3$ ✓.
2. $\lambda = 5$: $A - 5I = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Only one independent equation: $v_1 = v_2$, with v_3 free. So $E_5 = \operatorname{span}\big((1, 1, 0), (0, 0, 1)\big)$, dimension $2 = m_5$ ✓. (Rank check: $\operatorname{rank}(A - 5I) = 1 = 3 - m_5$.)

3. Diagonalisable. $P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ (columns $v_1 = (1,1,0)$, $v_2 = (0,0,1)$ for $\lambda = 5$, then $v_3 = (1,-1,0)$ for $\lambda = 3$), and $D = \operatorname{diag}(5,5,3)$.
4. Check: $Av_1 = (5,5,0) = 5v_1$, $Av_2 = (0,0,5) = 5v_2$, $Av_3 = (3,-3,0) = 3v_3$ ✓. So $A = PDP^{-1}$.

EXAMPLE // Worked example 3: repeated eigenvalue, NOT diagonalisable

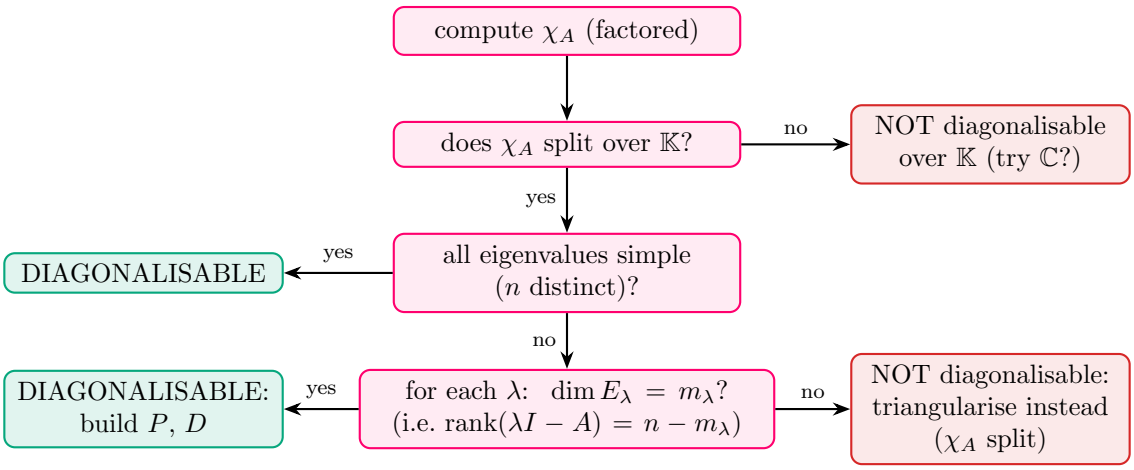
$A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$.

1. $\operatorname{tr} A = 4$, $\det A = 3 + 1 = 4$: $\chi_A = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$. Split, $m_2 = 2$.
2. $A - 2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, rank 1, so $\dim E_2 = 2 - 1 = 1 < 2 = m_2$. Concretely, $E_2 = \operatorname{span}\big((1,-1)\big)$.
3. **Not diagonalisable** (one eigenvalue, $A \neq 2I$, the same conclusion as the previous theorem). Triangularised in Section 7.

EXAMPLE // Worked example 4: depends on the field

The rotation by 90°, $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: $\chi_R = \lambda^2 + 1$.

- Over $\mathbb{R}$: no real root, χ_R doesn't split: **not diagonalisable** (no direction is kept by a quarter-turn).
- Over $\mathbb{C}$: $\chi_R = (\lambda - i)(\lambda + i)$, two distinct eigenvalues: **diagonalisable**, with $E_i = \operatorname{span}\big((1,-i)\big)$ and $E_{-i} = \operatorname{span}\big((1,i)\big)$. Check: $R(1,-i)^\top = (i,1)^\top = i(1,-i)^\top$ ✓.



5.4 What diagonalisation buys you: powers

THEOREM / PROPOSITION // Powers of a diagonalisable matrix

If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then for all $k \in \mathbb{N}$ (and $k \in \mathbb{Z}$ if all $\lambda_i \neq 0$):

$$A^k = PD^kP^{-1} = P\ \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)\, P^{-1}.$$

Each A^k is diagonalisable, with the same eigenvectors and eigenvalues λ_i^k. (If some $\lambda_i = 0$, then

0 is an eigenvalue of A^k; $A^0 = I_n$ has all eigenvalues equal to 1.)

RECEIPTS (proof) // domino cancellation

$A^2 = PDP^{-1}PDP^{-1} = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}$, then induction. Powers of a diagonal matrix are entry-wise. ■ *(and that's on periodt)*

EXAMPLE // Fibonacci via diagonalisation

$F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$: $0, 1, 1, 2, 3, 5, 8, 13, \ldots$ Write it as a linear system:

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}}_{A} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

1. $\chi_A = \lambda^2 - \lambda - 1$, roots $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ (the golden ratio) and $\psi = \frac{1-\sqrt{5}}{2} \approx -0.618$. Distinct: diagonalisable.
2. Eigenvectors: $(A - \lambda I)v = 0$ gives $v_1 = \lambda v_2$ (from the second row $v_1 - \lambda v_2 = 0$), so $v_\varphi = (\varphi, 1)$, $v_\psi = (\psi, 1)$.
3. Decompose the start: $(1, 0) = \frac{1}{\sqrt{5}}(v_\varphi - v_\psi)$, since $\varphi - \psi = \sqrt{5}$ and $1 - 1 = 0$.
4. Apply A^n: $A^n(1, 0) = \frac{1}{\sqrt{5}}(\varphi^n v_\varphi - \psi^n v_\psi)$. Read the second coordinate:

$$\boxed{F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}} \quad \text{(Binet's formula)}.$$

5. Check $n = 10$: $\varphi^{10} \approx 122.99$, $\psi^{10} \approx 0.008$, so $F_{10} \approx 122.98/2.236 \approx 55$ ✓ (and indeed $A^{10} = \begin{pmatrix} 89 & 55 \\ 55 & 34 \end{pmatrix}$).

Since $|\psi| < 1$, $\psi^n \to 0$: $F_n \approx \varphi^n/\sqrt{5}$ grows like φ^n. The *spectral radius* φ controls the growth: exactly the ML side quest of Section 3.

6 Polynomials of matrices and Cayley–Hamilton

TL;DR // Chapter 6 in one breath

You can plug a matrix into a polynomial: $P(A) = a_0 I + a_1 A + \cdots + a_m A^m$. If $P(A) = 0$, P is **annihilating** and every eigenvalue is a root of P. **Cayley–Hamilton**: the characteristic polynomial always annihilates: $\chi_A(A) = 0$. That lets you write A^{-1} and any A^m using only $I, A, \ldots, A^{n-1}$.

6.1 Annihilating polynomials

DEFINITION // Polynomial of a matrix, annihilating polynomial

If $P(X) = a_0 + a_1 X + \cdots + a_m X^m \in \mathbb{K}[X]$ and $A \in \mathcal{M}_n(\mathbb{K})$, then $P(A) = a_0 I_n + a_1 A + \cdots + a_m A^m$ (the constant becomes $a_0 I_n$!). Same for $f \in \mathcal{L}(E)$, with $f^0 = \mathrm{Id}_E$ and $f^k = f \circ \cdots \circ f$.
P is an **annihilating polynomial** of f if $P(f) = 0_{\mathcal{L}(E)}$ (the zero map).

THEOREM / PROPOSITION // An annihilating polynomial always exists

If $\dim E = n$, every $f \in \mathcal{L}(E)$ has a non-zero annihilating polynomial.

RECEIPTS (proof) // too many vectors

$\dim \mathcal{L}(E) = n^2$, and $\mathrm{Id}_E, f, f^2, \ldots, f^{n^2}$ are n^2+1 elements of it: they are linearly dependent. So there are $a_0, \ldots, a_{n^2}$, not all zero, with $a_0 \mathrm{Id} + a_1 f + \cdots + a_{n^2} f^{n^2} = 0$. Then $P = \sum a_k X^k \neq 0$ and $P(f) = 0$. (Cayley–Hamilton will give one of degree only n.) ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Eigenvalues are roots of every annihilating polynomial

If $P(f) = 0$, then every eigenvalue of f is a root of P: $\mathrm{Sp}(f) \subseteq \{z \in \mathbb{K} \mid P(z) = 0\}$.

RECEIPTS (proof) //

Let $f(v) = \lambda v$ with $v \neq 0$. By induction $f^k(v) = \lambda^k v$, so $P(f)(v) = \sum_k a_k f^k(v) = \sum_k a_k \lambda^k v = P(\lambda)v$. But $P(f) = 0$, so $P(\lambda)v = 0$, and $v \neq 0$ forces $P(\lambda) = 0$. ■ *(and that's on periodt)*

COMMON L (don't do this) // Inclusion, not equality

Roots of an annihilating polynomial are *candidates*, not necessarily eigenvalues. I_2 is annihilated by $P = (X-1)(X-7)$ (since $(I-I)(I-7I) = 0$), but 7 is not an eigenvalue of I_2.

EXAMPLE // Projections and symmetries

1. If $A^2 = A$ (a *projection*), $X^2 - X = X(X-1)$ annihilates A: $\mathrm{Sp}(A) \subseteq \{0, 1\}$.
2. If $A^2 = I$ (a *symmetry*, e.g. the transpose map of Lecture 3, Ex. 4): $\mathrm{Sp}(A) \subseteq \{-1, 1\}$.
3. If $N^k = 0$ (*nilpotent*): X^k annihilates N, so $\mathrm{Sp}(N) \subseteq \{0\}$.

DEFINITION // Minimal polynomial

The **minimal polynomial** μ_A of A is the unique **monic** polynomial of **smallest degree** with $\mu_A(A) = 0$. It divides every annihilating polynomial of A (in particular $\mu_A \mid \chi_A$, by Cayley–Hamilton), and all eigenvalues of A are roots of μ_A.

EXAMPLE // Minimal vs characteristic

$B = \mathrm{diag}(1, 2, 2)$: $\chi_B = (X-1)(X-2)^2$. But $(B-I)(B-2I) = \mathrm{diag}(0,1,1)\,\mathrm{diag}(-1,0,0) = 0$, while $B - I \neq 0$ and $B - 2I \neq 0$. So $\mu_B = (X-1)(X-2)$, of degree $2 < 3$.

THEOREM / PROPOSITION // Bonus: diagonalisability via the minimal polynomial

A is diagonalisable over $\mathbb{K} \iff \mu_A$ splits over $\mathbb{K}$ with **simple roots** $\iff$ A is annihilated by some split polynomial with simple roots. (Stated for culture; the proof uses the kernel lemma, beyond these notes.)

Quick use: $A^2 = A \Rightarrow X(X-1)$ annihilates A, split with simple roots $\Rightarrow$ every projection is diagonalisable. Same for $A^2 = I$ (over $\mathbb{R}$, roots ± 1).

6.2 Cayley–Hamilton

THEOREM / PROPOSITION // Cayley–Hamilton

For every $A \in \mathcal{M}_n(\mathbb{K})$ with $\chi_A(\lambda) = \det(\lambda I_n - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$:

$$\chi_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

Every matrix is a root of its own characteristic polynomial.

COMMON L (don't do this) // The fake proof

"$\chi_A(A) = \det(AI - A) = \det(0) = 0$" is **wrong**: $\chi_A(A)$ is a *matrix*, while $\det(\dots)$ is a *number*, and you can't substitute a matrix for λ inside a determinant like that. Cayley–Hamilton is a real theorem.

RECEIPTS (proof) // proof over $\mathbb{C}$, via triangularisation (next section)

Over $\mathbb{C}$, $A = PTP^{-1}$ with T upper triangular, diagonal $\lambda_1, \dots, \lambda_n$, and $\chi_A = \chi_T = \prod_i (X - \lambda_i)$. Since $\chi_A(A) = P\chi_A(T)P^{-1}$, it suffices to show $\chi_A(T) = (T - \lambda_1 I)(T - \lambda_2 I)\cdots(T - \lambda_n I) = 0$. Let $V_k = \operatorname{span}(e_1, \dots, e_k)$ ($V_0 = \{0\}$). T upper triangular means $Te_k \in V_k$, with k-th coordinate λ_k; so $(T - \lambda_k I)e_k \in V_{k-1}$, and $(T - \lambda_k I)V_k \subseteq V_{k-1}$. (Each factor $T - \lambda_j I$ also maps every V_k into itself.)
Claim: $(T - \lambda_1 I)\cdots(T - \lambda_k I)$ kills V_k. True for $k = 1$. If true for $k - 1$: apply the factors right to left. $T - \lambda_k I$ sends V_k into V_{k-1}, which the remaining $(T - \lambda_1 I)\cdots(T - \lambda_{k-1} I)$ kills. (The factors commute, being polynomials in T, so their order doesn't matter.) For $k = n$: the product kills $V_n = \mathbb{C}^n$, so it's 0. A real matrix is also a complex one, so the theorem holds over $\mathbb{R}$ too. ■ *(and that's on periodt)*

EXAMPLE // Checking Cayley–Hamilton

$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, $\chi_A = \lambda^2 - 5\lambda + 6$. $A^2 = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix}$, and

$$A^2 - 5A + 6I_2 = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 10 & 5 \\ 0 & 15 \end{pmatrix} + \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \checkmark$$

6.3 What Cayley–Hamilton is good for

EXAM CHEAT CODE // "Compute A^{-1} with Cayley–Hamilton"

1. Compute $\chi_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$. Note $c_0 = (-1)^n \det A$, so A invertible $\iff c_0 \neq 0$.
2. Write $\chi_A(A) = 0$ and move $c_0 I$ to the other side: $A\,(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I) = -c_0 I$.
3. Divide: $A^{-1} = -\dfrac{1}{c_0}\big(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I\big)$.

EXAMPLE // Inverse of a 2×2 (a classic exercise)

For any invertible 2×2: $A^2 - \operatorname{tr}(A)A + \det(A)I = 0$, so $A(\operatorname{tr}(A)I - A) = \det(A)I$ and

$$A^{-1} = \frac{1}{\det A}\big(\operatorname{tr}(A)\, I - A\big).$$

With $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\operatorname{tr}(A)I - A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. That's the Lecture 2 formula, derived a third way.

For $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$: $\det = 6$, $\operatorname{tr} = 5$, so $A^{-1} = \frac{1}{6}\begin{pmatrix} 3 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix}$. Check: $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{3} + \frac{1}{3} \\ 0 & 1 \end{pmatrix} = I$ ✓.

EXAMPLE // Inverse of a 3×3

$M = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$.

1. $\chi_M(\lambda) = \lambda^3 - 4\lambda^2 + 5\lambda - 3$ ($\operatorname{tr} M = 4$ and $\det M = 3$, consistent with $c_2 = -4$, $c_0 = -3$).
2. Cayley–Hamilton: $M^3 - 4M^2 + 5M - 3I = 0$, so $M(M^2 - 4M + 5I) = 3I$.
3. $M^2 = \begin{pmatrix} 4 & 3 & 1 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$, so $M^{-1} = \frac{1}{3}(M^2 - 4M + 5I) = \frac{1}{3}\begin{pmatrix} 4-8+5 & 3-4 & 1 \\ 1 & 1-4+5 & 2-4 \\ 3-4 & 1 & 1-4+5 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & -1 & 1 \\ 1 & 2 & -2 \\ -1 & 1 & 2 \end{pmatrix}$.
4. Check row 1 of MM^{-1}: $\frac{1}{3}(2\cdot 1 + 1\cdot 1,\ 2\cdot(-1) + 1\cdot 2,\ 2\cdot 1 + 1\cdot(-2)) = (1, 0, 0)$ ✓.

TL;DR // Practical uses of Cayley–Hamilton

- **Reduce powers:** any A^m ($m \geq n$) is a combination of $I, A, \ldots, A^{n-1}$ (divide X^m by χ_A and keep the remainder).
- **Inverses:** polynomial formula for A^{-1} when $\det A \neq 0$.
- **Minimal polynomial:** μ_A divides χ_A.
- **Linear recurrences:** $x_k = A^k x_0$ satisfies a recurrence whose coefficients are those of χ_A (control theory… and deep learning).

7 Triangularisation and nilpotent matrices

TL;DR // Chapter 7 in one breath

Not diagonalisable? If χ_A splits (always over $\mathbb{C}$), you can still make A **upper triangular**: $A = PTP^{-1}$. Then $T = D + N$ with D diagonal and N nilpotent ($N^k = 0$), and powers are computed with the binomial formula, which stops after a few terms.

7.1 Triangularisability

DEFINITION // Triangularisable

$f \in \mathcal{L}(E)$ is **triangularisable** if some basis $\mathcal{B}$ makes $[f]_{\mathcal{B}}$ (upper) triangular. $A \in \mathcal{M}_n(\mathbb{K})$ is triangularisable if $A = PTP^{-1}$ with P invertible and T triangular. Every diagonalisable matrix is triangularisable (diagonal is triangular), **not** conversely.

THEOREM / PROPOSITION // Reading triangularity on the basis

For a basis $\mathcal{B} = (e_1, \dots, e_n)$: $[f]_{\mathcal{B}}$ is upper triangular $\iff$ for all i, $f(e_i) \in \operatorname{span}(e_1, \dots, e_i)$. (Column i holds the coordinates of $f(e_i)$; "zeros below the diagonal" means no component along $e_{i+1}, \dots, e_n$.) In particular e_1 is an eigenvector.

THEOREM / PROPOSITION // Triangularisability criterion

f (or A) is triangularisable over $\mathbb{K}$ $\iff$ χ_f (or χ_A) splits over $\mathbb{K}$. In particular, **every complex matrix is triangularisable**; a real matrix is triangularisable over $\mathbb{R}$ iff all its eigenvalues are real.

RECEIPTS (proof) // induction on n

($\Rightarrow$) If $A = PTP^{-1}$ with T triangular, $\chi_A = \chi_T = \prod(X - t_{ii})$ splits.
($\Leftarrow$) $n = 1$ is trivial. Assume the result in dimension $n-1$, and let χ_A split. It has a root λ_1, an eigenvalue with eigenvector u_1. Complete u_1 into a basis; in it, $A \sim \begin{pmatrix} \lambda_1 & * \\ 0 & B \end{pmatrix}$ with $B \in \mathcal{M}_{n-1}(\mathbb{K})$ and $\chi_A = (X - \lambda_1)\chi_B$. So χ_B splits too, and by induction $B = QT'Q^{-1}$ with T' triangular. Then conjugating by $\begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}$ turns $\begin{pmatrix} \lambda_1 & * \\ 0 & B \end{pmatrix}$ into $\begin{pmatrix} \lambda_1 & * \\ 0 & T' \end{pmatrix}$, which is triangular.
■ *(and that's on periodt)*

THEOREM / PROPOSITION // Trace, determinant and multiplicities

If $\chi_f(X) = \prod_{\lambda \in \operatorname{Sp}(f)} (X - \lambda)^{m_\lambda}$ is split, then

$$\sum_{\lambda} m_\lambda = n, \qquad \det(f) = \prod_{\lambda} \lambda^{m_\lambda}, \qquad \operatorname{tr}(f) = \sum_{\lambda} m_\lambda \lambda.$$

RECEIPTS (proof) //

Triangularise: T has each λ exactly m_λ times on its diagonal ($\chi_T = \prod(X - t_{ii})$). Similar matrices share det and tr; for triangular T, $\det T = \prod t_{ii}$ and $\operatorname{tr} T = \sum t_{ii}$. ■ *(and that's on periodt)*

7.2 Nilpotent matrices

DEFINITION // Nilpotent, index

$N \in \mathcal{M}_n(\mathbb{K})$ is **nilpotent** if $N^k = 0$ for some $k \geq 1$. The smallest such k is the **index** of N.

THEOREM / PROPOSITION // Facts about nilpotent matrices

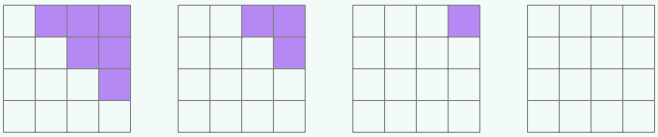
1. An upper triangular matrix with **zeros on the diagonal** is nilpotent, of index $\leq n$.
2. If N is nilpotent: $\mathrm{Sp}(N) = \{0\}$, $\chi_N = X^n$, $\det N = 0$, $\operatorname{tr} N = 0$. So N is never invertible.
3. A nilpotent $N \neq 0$ is never diagonalisable.

RECEIPTS (proof) //

(1) With $V_k = \mathrm{span}(e_1, \ldots, e_k)$, a strictly upper triangular N sends V_k into V_{k-1}. So N^k sends $\mathbb{K}^n = V_n$ into V_{n-k}, and $N^n = 0$. (2) X^k annihilates N, so every eigenvalue is a root of X^k: $\mathrm{Sp}(N) \subseteq \{0\}$. Over $\mathbb{C}$, χ_N splits with only the root 0: $\chi_N = X^n$; its coefficients give $\operatorname{tr} N = 0$ and $\det N = 0$. (3) Diagonalisable with single eigenvalue 0 would force $N = 0 \cdot I = 0$. ■ *(and that's on periodt)*

EXAMPLE // Watch the non-zero part climb

$$B = \begin{pmatrix} 0 & 2 & 1 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}: \quad B^2 = \begin{pmatrix} 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B^3 = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B^4 = 0. \text{ Index 4.}$$



B B^2 B^3 $B^4 = 0$

Each power pushes the non-zero diagonals one step further up-right, until they fall off the matrix.

EXAMPLE // $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$N^2 = 0$, index 2. The simplest nilpotent matrix, and the simplest non-diagonalisable one.

7.3 Powers of non-diagonalisable matrices: $T = D + N$

THEOREM / PROPOSITION // Binomial formula when matrices commute

If D and N commute ($DN = ND$) and $N^r = 0$, then

$$(D+N)^m = \sum_{k=0}^{r-1} \binom{m}{k} D^{m-k} N^k \qquad \text{(the sum stops at } k = r-1\text{)}.$$

In particular, for $D = \lambda I$ (which commutes with everything): $(\lambda I + N)^m = \sum_{k=0}^{r-1} \binom{m}{k} \lambda^{m-k} N^k$.

COMMON L (don't do this) // Commuting is required

$(A+B)^2 = A^2 + AB + BA + B^2$. Only when $AB = BA$ does this become $A^2 + 2AB + B^2$, and only then does the binomial formula hold. λI commutes with every matrix, which is why the formula is so often used with $D = \lambda I$.

EXAMPLE // Jordan block powers

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = I_3 + N \text{ with } N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},\ N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ N^3 = 0.$$

1. Cayley–Hamilton: $\chi_J = (\lambda - 1)^3$, so $(J - I)^3 = 0$, i.e. $J^3 - 3J^2 + 3J - I = 0$.
2. I and N commute, so $J^m = (I + N)^m = I + mN + \binom{m}{2}N^2$ (the other terms vanish).
3. $J^m = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$. For $m = 5$: $\begin{pmatrix} 1 & 5 & 10 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$ ✓.

Cayley–Hamilton + nilpotent part = powers of a matrix even when it's not diagonalisable.

7.4 The method

EXAM CHEAT CODE // "Triangularise A" (when χ_A splits)

Let $\chi_A = \prod_{i=1}^{p}(X - \lambda_i)^{m_i}$ with distinct λ_i and $\sum m_i = n$.

1. For each λ_i, find a basis $\mathcal{B}_i$ of $\ker(\lambda_i I - A)$.
2. If $\mathrm{Card}(\mathcal{B}_i) = m_i$ for all i: diagonalise instead (Section 5).
3. Otherwise, complete $\mathcal{B}_i$ with **generalised eigenvectors**: vectors of $\ker\left((A-\lambda_i I)^2\right)$, then $\ker\left((A-\lambda_i I)^3\right)$, …, until you have m_i vectors. In practice: solve $(A - \lambda_i I)w = v$ where v is an eigenvector already found, then $(A - \lambda_i I)w' = w$, etc. (a *chain*).
4. $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p$ (in chain order: eigenvector first) is a basis in which A is upper triangular. P = these vectors as columns, $T = P^{-1}AP$.
5. Check $AP = PT$.

EXAMPLE // Triangularising $A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$ (worked example 3, continued)

1. $\chi_A = (\lambda - 2)^2$, $E_2 = \mathrm{span}(v_1)$ with $v_1 = (1, -1)$: one vector short.
2. Solve $(A - 2I)w = v_1$: $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, i.e. $w_1 + w_2 = 1$. Take $w = v_2 = (1, 0)$.
3. Then $Av_1 = 2v_1$ and $Av_2 = v_1 + 2v_2$. In the basis (v_1, v_2): $T = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, with $P = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$.
4. Check $AP = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -2 & -1 \end{pmatrix}$ and $PT = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -2 & -1 \end{pmatrix}$ ✓.

5. Bonus: $T = 2I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so $T^m = 2^m I + m2^{m-1}N$ and $A^m = PT^mP^{-1}$.

8 The big worked exercise (putting it all together)

Let $a > 0$ and $A = \begin{pmatrix} -1 & a & -a \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$. (1) Compute χ_A and show A is not diagonalisable. (2) Find v_1, v_2, v_3 with $Av_1 = -v_1$, $Av_2 = v_1 - v_2$, $Av_3 = v_1 + v_2 - v_3$. (3) Show A is similar to $T = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$. (4) Use a nilpotent matrix to compute A^n, $n \in \mathbb{N}$. (5) Can we compute A^n for $n \in \mathbb{Z}$?

SOLUTION // (1) Characteristic polynomial, non-diagonalisability

$\lambda I_3 - A = \begin{pmatrix} \lambda+1 & -a & a \\ -1 & \lambda+1 & 0 \\ -1 & 0 & \lambda+1 \end{pmatrix}$. Expand along the first row:

$$\begin{aligned} \chi_A(\lambda) &= (\lambda+1)\begin{vmatrix} \lambda+1 & 0 \\ 0 & \lambda+1 \end{vmatrix} - (-a)\begin{vmatrix} -1 & 0 \\ -1 & \lambda+1 \end{vmatrix} + a\begin{vmatrix} -1 & \lambda+1 \\ -1 & 0 \end{vmatrix} \\ &= (\lambda+1)^3 + a\big(-(\lambda+1)\big) + a(\lambda+1) = (\lambda+1)^3. \end{aligned}$$

Only eigenvalue: -1, with $m_{-1} = 3$. Now $A + I_3 = \begin{pmatrix} 0 & a & -a \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$: rows 2 and 3 give $v_1 = 0$, row 1 gives $a(v_2 - v_3) = 0$, so $v_2 = v_3$ (using $a \neq 0$). Hence $E_{-1} = \{(0,t,t)^\top\}$, $\dim E_{-1} = 1 < 3$: **not diagonalisable**. (Quicker: the only eigenvalue is -1 and $A \neq -I_3$.)

SOLUTION // (2) The vectors v_1, v_2, v_3 (a chain)

The relations say $(A+I)v_1 = 0$, $(A+I)v_2 = v_1$, $(A+I)v_3 = v_1 + v_2$. Write $N = A + I_3$.

1. $Nv_1 = 0$: take $v_1 = (0,1,1)^\top$ (from (1)).
2. $Nv_2 = v_1$: $N(x,y,z)^\top = (a(y-z),\ x,\ x)^\top = (0,1,1)^\top$ gives $x = 1$, $y = z$. Take $v_2 = (1,0,0)^\top$.
3. $Nv_3 = v_1 + v_2 = (1,1,1)^\top$: $x = 1$ and $a(y-z) = 1$. Take $v_3 = (1, \frac{1}{a}, 0)^\top$ (needs $a \neq 0$).

Check: $Av_2 = (-1,1,1)^\top = v_1 - v_2$ ✓; $Av_3 = (-1+1,\ 1-\frac{1}{a},\ 1)^\top = (0, 1-\frac{1}{a}, 1)^\top$ and $v_1 + v_2 - v_3 = (0+1-1,\ 1+0-\frac{1}{a},\ 1)^\top$ ✓.

SOLUTION // (3) Similarity to T

Let $P = (v_1\ v_2\ v_3) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & \frac{1}{a} \\ 1 & 0 & 0 \end{pmatrix}$. It is invertible ($\det P = 1 \cdot \begin{vmatrix} 1 & 1 \\ 0 & \frac{1}{a} \end{vmatrix} = \frac{1}{a} \neq 0$, expanding along row 3). The relations of (2) say exactly that the columns of the matrix of A in the basis

(v_1, v_2, v_3) are $(-1,0,0)$, $(1,-1,0)$, $(1,1,-1)$:

$$P^{-1}AP = T = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}, \qquad A = PTP^{-1}.$$

SOLUTION // (4) A^n for $n \in \mathbb{N}$, via a nilpotent matrix

Write $A = -I_3 + N$ with $N = A + I_3 = \begin{pmatrix} 0 & a & -a \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

1. $N^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & a & -a \\ 0 & a & -a \end{pmatrix} \neq 0$ and $N^3 = 0$. (N is similar to $T + I$, strictly upper triangular, so this was expected.)
2. $-I_3$ and N commute, so the binomial formula applies and stops at $k = 2$:

$$A^n = (-I_3 + N)^n = \sum_{k=0}^{2} \binom{n}{k} (-1)^{n-k} N^k = (-1)^n \Big(I_3 - nN + \binom{n}{2} N^2 \Big).$$

3. Explicitly, with $\binom{n}{2} = \frac{n(n-1)}{2}$:

$$A^n = (-1)^n \begin{pmatrix} 1 & -na & na \\ -n & 1 + \binom{n}{2} a & -\binom{n}{2} a \\ -n & \binom{n}{2} a & 1 - \binom{n}{2} a \end{pmatrix}, \qquad n \in \mathbb{N}.$$

4. Checks: $n = 0$ gives I_3 ✓; $n = 1$ gives $-\begin{pmatrix} 1 & -a & a \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = A$ ✓.

SOLUTION // (5) A^n for $n \in \mathbb{Z}$

$\det A = \det T = (-1)^3 = -1 \neq 0$, so A is invertible and A^{-n} makes sense. $A^{-1} = (-I + N)^{-1} = -(I - N)^{-1}$, and since $N^3 = 0$:

$$(I - N)(I + N + N^2) = I - N^3 = I \quad \Longrightarrow \quad (I - N)^{-1} = I + N + N^2.$$

By induction (or the generalised binomial series, which stops because $N^3 = 0$), $(I - N)^{-n} = I + nN + \binom{n+1}{2} N^2$ for $n \in \mathbb{N}$. Hence

$$A^{-n} = (-1)^n \Big(I_3 + nN + \binom{n+1}{2} N^2 \Big) = (-1)^n \begin{pmatrix} 1 & na & -na \\ n & 1 + \binom{n+1}{2} a & -\binom{n+1}{2} a \\ n & \binom{n+1}{2} a & 1 - \binom{n+1}{2} a \end{pmatrix}, \qquad n \in \mathbb{N}.$$

Check $n = 1$: $A^{-1} = -\begin{pmatrix} 1 & a & -a \\ 1 & 1+a & -a \\ 1 & a & 1-a \end{pmatrix}$, and indeed $AA^{-1} = I_3$ (first row: $-(-1 + a - a,\ -a + a + a^2 - a^2,\ a - a^2 - a + a^2) = (1,0,0)$ ✓). Together with (4), A^m is known for every $m \in \mathbb{Z}$. (All formulas checked by computer algebra for $|m| \leq 5$.)

9 Practice round

Short exercises in the style of the exam, with solutions. Cover the solution, try, then check.

EXAMPLE // P1. Diagonalise $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and compute A^n

1. $\operatorname{tr} = 2$, $\det = 1 - 4 = -3$: $\chi_A = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$. Distinct: diagonalisable.
2. E_3: $A - 3I = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$, $v_1 = v_2$: $(1, 1)$. E_{-1}: $A + I = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$, $v_1 = -v_2$: $(1, -1)$.
3. $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \operatorname{diag}(3, -1)$, $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.
4. $A^n = PD^nP^{-1} = \frac{1}{2}\begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}$. Check $n = 1$: $\frac{1}{2}\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix} = A$ ✓.

EXAMPLE // P2. Is $C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ diagonalisable?

1. Upper triangular: $\chi_C = (\lambda - 2)(\lambda - 1)^2$, $m_2 = 1$, $m_1 = 2$.
2. $C - I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ has rank 2, so $\dim E_1 = 3 - 2 = 1 < 2 = m_1$.
3. **Not diagonalisable.** (It's block diagonal: [2] and a 2×2 Jordan block.)

EXAMPLE // P3. True or false?

1. *"A and $A^\top$ have the same eigenvalues."* **True**: $\chi_{A^\top}(\lambda) = \det(\lambda I - A^\top) = \det\left((\lambda I - A)^\top\right) = \chi_A(\lambda)$. (The eigen*vectors* can differ.)
2. *"If A is invertible and diagonalisable, so is A^{-1}."* **True**: $A^{-1} = PD^{-1}P^{-1}$, same eigenvectors, eigenvalues $1/\lambda_i$.
3. *"A real matrix with real eigenvalues is diagonalisable."* **False**: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. (It is *triangularisable* though.)
4. *"If $A^2 = 0$ then $A = 0$."* **False**: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. But $A^2 = 0$ and A diagonalisable $\Rightarrow A = 0$.
5. *"Similar matrices have the same eigenvectors."* **False**: same χ, same eigenvalues, but $B = P^{-1}AP$ has eigenvectors $P^{-1}v$.
6. *"A symmetric real matrix is diagonalisable."* **True**, and even with an orthonormal eigenbasis: that's the *spectral theorem*, next lecture.

EXAMPLE // P4. Eigenvalues without computing χ

$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ (J, the all-ones matrix, $n = 3$).

1. $\operatorname{rank} A = 1$, so $\dim \ker A = 2$: 0 is an eigenvalue with $\dim E_0 = 2$, hence $m_0 \geq 2$.

2. $A(1,1,1)^\top = (3,3,3)^\top$: 3 is an eigenvalue.
3. $\operatorname{tr} A = 3 = 0+0+3$ ✓: $\chi_A = \lambda^2(\lambda-3)$, $\dim E_0 = 2 = m_0$, $\dim E_3 = 1 = m_3$: diagonalisable.

10 Final boss: the cheat sheet

10.1 "If they ask you this, do that"

If asked...	**Do this**
Is v an eigenvector?	Compute Av; check $v \neq 0$ and $Av = \lambda v$ for some λ.
Eigenvalues	Roots of $\chi_A = \det(\lambda I - A)$. Triangular: diagonal. Block: union of blocks. 2×2: $\lambda^2 - \operatorname{tr}\lambda + \det$.
Eigenspace E_λ	Solve $(A-\lambda I)v = 0$ by elimination; $\dim E_\lambda = n - \operatorname{rank}(A-\lambda I)$.
Is 0 an eigenvalue?	$\iff \det A = 0 \iff A$ not invertible.
Diagonalisable?	χ_A split AND $\dim E_\lambda = m_\lambda$ for all λ. Shortcut: n distinct eigenvalues. One eigenvalue: only if $A = \lambda I$.
Diagonalise	P = eigenvectors (columns), D = eigenvalues in the same order. Check $AP = PD$.
A^k (diagonalisable)	PD^kP^{-1}.
A^k (not diagonalisable)	Triangularise $A = PTP^{-1}$, $T = D+N$ with $DN = ND$ (e.g. $D = \lambda I$), binomial formula stops at N^{r-1}.
A^{-1} by Cayley–Hamilton	$\chi_A(A) = 0$; isolate the constant term $c_0 I$; divide by $-c_0$.
Show $\operatorname{Sp}(A) \subseteq S$	Find an annihilating polynomial P ($A^2 = A$, $A^2 = I$, $N^k = 0\ldots$); eigenvalues are roots of P.
Triangularise	Eigenvectors, then chains $(A-\lambda I)w = v$ for missing vectors.
Nilpotent?	$N^k = 0$; then $\operatorname{Sp} = \{0\}$, $\operatorname{tr} = \det = 0$, not invertible, not diagonalisable unless 0.
Long-term behaviour of $x_{t+1} = Ax_t$	Eigen-decompose x_0; components with $\lvert\lambda\rvert < 1$ die, $\lvert\lambda\rvert > 1$ explode; look at $\rho(A)$.

10.2 Formula sheet

Eigen	$Av = \lambda v$, $v \neq 0$; $E_\lambda = \ker(A-\lambda I)$; $\operatorname{Sp}(A)$; $\rho(A) = \max\lvert\lambda\rvert$
Char. polynomial	$\chi_A = \det(\lambda I - A) = \lambda^n - \operatorname{tr}(A)\lambda^{n-1} + \cdots + (-1)^n \det A$
2×2	$\chi_A = \lambda^2 - \operatorname{tr}(A)\lambda + \det A$; $A^{-1} = \frac{1}{\det A}(\operatorname{tr}(A)I - A)$
Multiplicities	$1 \leq \dim E_\lambda \leq m_\lambda$; split $\Rightarrow \sum m_\lambda = n$, $\operatorname{tr} = \sum m_\lambda\lambda$, $\det = \prod \lambda^{m_\lambda}$
Shifts etc.	$A+\gamma I$: $\lambda+\gamma$; A^{-1}: $1/\lambda$; A^k: λ^k; $P(A)$: $P(\lambda)$
Diagonalisation	$A = PDP^{-1} \iff AP = PD$; $A^k = PD^kP^{-1}$
Criterion	diag. $\iff \chi_A$ split and $\dim E_\lambda = m_\lambda\ \forall\lambda$; triang. $\iff \chi_A$ split
Cayley–Hamilton	$\chi_A(A) = 0$; $\mu_A \mid \chi_A$; $P(A) = 0 \Rightarrow \operatorname{Sp}(A) \subseteq$ roots of P
Nilpotent	$N^k = 0$; $(\lambda I + N)^m = \sum_{k<r} \binom{m}{k} \lambda^{m-k} N^k$

10.3 Hall of Ls

COMMON L (don't do this) // Do not

1. Call 0 an eigenvector. (Eigenvalue 0: fine. Eigenvector 0: never.)
2. Conclude "repeated eigenvalue ⇒ not diagonalisable". Check $\dim E_\lambda$ (I_n is diagonal!).
3. Conclude "real eigenvalues ⇒ diagonalisable". $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
4. Write DP instead of PD, or put eigenvalues in D in a different order from the eigenvectors in P.
5. Forget to check that χ_A *splits* over the field you're working in ($\mathbb{R}$ vs $\mathbb{C}$).
6. "Prove" Cayley–Hamilton by plugging $\lambda = A$ into the determinant.
7. Use the binomial formula on non-commuting matrices.
8. Write $P(A) = a_0 + a_1 A + \dots$ (a number plus matrices). It's $a_0 I + a_1 A + \dots$
9. Assume every root of an annihilating polynomial is an eigenvalue. Only the inclusion holds.
10. Mix $\det(\lambda I - A)$ and $\det(A - \lambda I)$ inside one computation (they differ by $(-1)^n$).

10.4 The whole lecture in 10 lines

TL;DR // Screenshot this

1. Eigenvector: $v \neq 0$ with $Av = \lambda v$; the direction is kept, the length is scaled by λ.
2. $E_\lambda = \ker(A - \lambda I)$; eigenspaces of distinct eigenvalues are in direct sum, so at most n eigenvalues.
3. $\chi_A = \det(\lambda I - A)$: monic, degree n, roots = eigenvalues, carries tr and det, similarity-invariant.
4. $1 \leq \dim E_\lambda \leq m_\lambda$.
5. Diagonalisable $\iff$ eigenbasis $\iff$ χ_A split and $\dim E_\lambda = m_\lambda$. n distinct eigenvalues suffice.
6. $A = PDP^{-1}$, $AP = PD$, $A^k = PD^kP^{-1}$.
7. Annihilating polynomial $P(A) = 0 \Rightarrow$ eigenvalues are roots of P. Cayley–Hamilton: $\chi_A(A) = 0$ (inverses, powers).
8. Triangularisable $\iff$ χ_A split (always over $\mathbb{C}$); then $\mathrm{tr} = \sum \lambda$, $\det = \prod \lambda$.
9. Nilpotent: $N^k = 0$, $\mathrm{Sp} = \{0\}$. Powers of $\lambda I + N$ by the binomial formula.
10. ML: stability of $x_{t+1} = Ax_t$ and RNN gradients via $\rho(A)$; Markov chains and PageRank via the eigenvalue 1.

Want more practice?

Review the method, and if solving the linear systems for eigenspaces is still slow, redo Example 2.2 of *Mathematics for Machine Learning*. Next class: inner product spaces, normed spaces, the spectral

theorem (prepare with MML, Chapter 4).

Next episode: inner products, norms and the spectral theorem. Stay tuned, bestie.